1. You are given the following matrix A and its reduced row echelon form R.

$$A = \begin{bmatrix} -1 & 0 & -1 & 2 \\ -3 & 1 & -1 & 13 \\ -1 & 2 & 0 & 10 \\ -1 & 1 & -1 & 5 \\ 3 & 0 & 2 & -8 \end{bmatrix} \qquad R = \begin{bmatrix} 1 & 0 & 0 & -4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(a) (4 points) Write the general solution to $A\mathbf{x} = \mathbf{0}$ in parametric vector form.

(b) (2 points) Find the specific solution to $A\mathbf{x} = \mathbf{0}$ for which $x_1 + 6 = x_2$.

(c) (2 points) It is given that $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ is a solution to $A\mathbf{x} = \mathbf{c}$. Find $\mathbf{c}$.

(d) (1 point) Find the dimension of null(A).

(e) (2 points) What is the rank of A?

(f) (2 points) Let $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$, and $\mathbf{a}_4$ denote the columns of A. Write the first column of A as a linear combination of the other three columns of A.

2. Let $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 1 & 2 \\ 3 & 2 & 3 \end{bmatrix}$

(a) (4 points) Find the inverse of A.

(b) (3 points) Use your answer in part (a) to find the matrix B, if $BA + A^T = \begin{bmatrix} 0 & 2 & 1 \\ 2 & -7 & -1 \\ 7 & -5 & 3 \end{bmatrix}$.

3. (4 points) Find the only value of a such that the following matrix equation has infinitely many solutions. $\begin{bmatrix} 1 & a-4 \\ a & a-6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -a-1 \\ -12 \end{bmatrix}$

4. Let A, B, and C be 3×3 matrices such that $\det(ABC) = 0$, $\det(2B^TC) = 6$, $\det(C^{-1}) = 2$.

(a) (2 points) Find $\det(A)$.

(b) (2 points) Find $\det(B)$.

(c) (2 points) Find $\det(C)$.

5. (5 points) The determinant of the following matrix A is 2026. Find k.

$$A = \begin{bmatrix} 1 & 0 & 3 & 1 \\ 2 & 0 & 7 & 2 \\ 3 & k & 8 & 11 \\ 3 & 0 & 2 & 5 \end{bmatrix}$$

6. (4 points) You are given $A = \begin{bmatrix} 7 & -5 & 0 & 4 \\ -3 & 3 & -6 & 9 \\ 5 & 1 & 1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 & 2 & -3 \\ 7 & -5 & 0 & 4 \\ 5 & 1 & 1 & -1 \end{bmatrix}$

Find two elementary matrices E_1 and E_2 such that $B = E_2E_1A$. (There is more than one correct answer, but you only need to find one pair of matrices that work.)

7. Let $A = \begin{bmatrix} 3 & 1 & -1 \\ -4 & -2 & 1 \\ 2 & 0 & -1 \end{bmatrix}$.

(a) (4 points) Find the characteristic polynomial of A.

(b) (3 points) Find all eigenvalues of A. (You do not need to find eigenvectors.)

8. (6 points) It is given that $A = \begin{bmatrix} -1 & -4 & 4 \\ 0 & 1 & 0 \\ -2 & -4 & 5 \end{bmatrix}$ has two eigenvalues: $\lambda_1 = 1$ and $\lambda_2 = 3$. Find a diagonal matrix D and an invertible matrix P such that $AP = PD$.

9. (4 points) Let $A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 2 & 1 & 4 & -4 \\ 1 & 4 & 2 & 2 \\ 0 & 3 & 0 & 2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$. It is given that $\det(A) = 10$. Using Cramer's Rule, find x_4 only in the solution to $A\mathbf{x} = \mathbf{b}$.

10. (3 points) Let $H = \{\mathbf{x} \in \mathbb{R}^3 : x_1 \leqslant x_2 \leqslant x_3\}$. Show that H is **not** a subspace of $\mathbb{R}^3$.

11. (5 points) Let $H = \left\{ \mathbf{x} \in \mathbb{R}^4 : \mathbf{x} \text{ is orthogonal to } \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \right\}$. Find a basis for H.

12. (6 points) Provide specific examples of the following:

(a) Vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^3$ such that $\text{span}\{\mathbf{u}, \mathbf{v}\}$ is not a plane.

(b) A 2×2 matrix A such that $A \neq 0$ but $A^2 = 0$.

(c) Vectors $\mathbf{x}$, $\mathbf{y}$, $\mathbf{z}$ in $\mathbb{R}^3$ such that $\{\mathbf{x}, \mathbf{y}\}$ and $\{\mathbf{y}, \mathbf{z}\}$ are linearly dependent sets, but $\{\mathbf{x}, \mathbf{z}\}$ is a linearly independent set.

13. (6 points) Let $\mathcal{L}$ be the line $\mathbf{x} = \begin{bmatrix} 5 \\ 4 \\ 1 \end{bmatrix} + t \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ for $t \in \mathbb{R}$. Let P be the point $(-2, 7, 0)$. Find the point on the line $\mathcal{L}$ that is nearest to point P.

14. You are given the following four points: $P(-3, -3, 8)$, $Q(0, -5, 6)$, $R(1, -4, 2)$, and $S(4, -1, 5)$.

(a) (2 points) Find a parametric vector equation of the line containing P and Q.

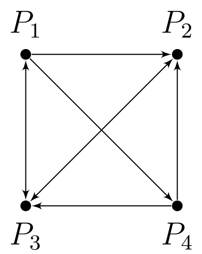
(b) (4 points) Find an equation of the plane containing P and Q that is parallel to the line containing R and S. Your answer must be in the form $ax + by + cz = d$.

(c) (2 points) Find the area of the triangle whose vertices are P, Q, and R.

(d) (2 points) Find the cosine of the angle formed by vector $\overrightarrow{PQ}$ and vector $\overrightarrow{PS}$.

15. You are given the following directed graph.

 Question 15 continues on the next page.



(a) (4 points) Write the adjacency matrix M for the graph.

(b) The third row of M^4 is provided below.

$$M^4 = \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ 2 & 4 & 5 & 2 \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

i. (2 points) How many **walks of length 4** start at P_3 and end at P_4?

ii. (2 points) How many **closed walks of length 5** start at P_3?

16. (6 points) The following diagram shows a bent metal bar and a cable (tied between A and B). The tension in the cable is 18N. The points shown are $A(3,4,7)$, $B(7,0,0)$, and $C(0,4,0)$, measured in meters. Find the torque exerted at A about C.